

Article

Assessment of Transitional RANS Models and Implementation of Transitional IDDES Method for Boundary Layer Transition and Separated Flows in OpenFOAM-V2312

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Abstract

Traditional hybrid RANS/LES methods often struggle to accurately capture both the boundary layer transition and flow separation simultaneously due to their reliance on fully turbulent RANS models. To address this limitation, the present study first evaluates three transitional RANS models (γ -Re $_{\theta}$ -SST, γ -SST, and K γ -SST) on the E387 airfoil. The results demonstrate that the γ -SST model offers the best balance of accuracy and computational efficiency in predicting laminar separation bubbles (LSBs) and transition points. Building on this, we implement the γ -SST-IDDES model into OpenFOAM-v2312, which integrates the γ -SST transitional RANS model with the Improved Delayed Detached Eddy Simulation (IDDES) approach. This coupling allows for the simultaneous prediction of the laminar-turbulent transition and high-fidelity resolution of separated flows. The γ -SST-IDDES model is rigorously validated across three airfoil cases with distinct separation characteristics: E387 (small separation), DBLN-526 (moderate separation), and NACA 0021 (massive separation). The results show that the γ -SST-IDDES model outperforms conventional methods, capturing leading-edge LSBs with high accuracy compared to fully turbulent IDDES. Additionally, it successfully resolves complex 3D vortical structures in separated regions, whereas unsteady URANS provides only quasi-2D results.

Keywords: laminar separation bubble; boundary layer transition; separated flow; γ -k ω SST-IDDES; OpenFOAM-v2312



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1. Introduction

Accurate prediction of boundary layer transition and separated flows is a critical challenge in computational fluid dynamics (CFD), particularly for aerodynamic applications such as wind turbine blades and aircraft wings. For example, modern wind turbine blades often employ blunt trailing-edge airfoils to enhance structural stability and aerodynamic performance. These airfoils are prone to complex flow phenomena, including boundary layer transition and massive flow separation, which significantly impact aerodynamic efficiency and load distributions. Traditional CFD methods, such as fully turbulent Reynolds-Averaged Navier–Stokes (RANS) models, fail to capture these phenomena due to their inherent assumption of a fully turbulent flow. While high-fidelity methods like Direct Numerical Simulation (DNS) and Large Eddy Simulation (LES) can resolve these flows,

their prohibitive computational cost makes them impractical for industrial applications, especially at high Reynolds numbers.

To address the limitations of fully turbulent RANS models in predicting the boundary layer transition, researchers have developed transitional RANS models that incorporate transition mechanisms into traditional turbulence frameworks. Among these, Local Correlation-based Transition Models (LCTMs) have gained widespread acceptance due to their ability to predict natural, bypass, and separation-induced transitions using local variables. A pioneering example is the γ - $Re_{\theta t}$ -SST model proposed by Menter et al. [1–4], which introduced transport equations for intermittency (γ) and the momentum-thickness Reynolds number ($Re_{\theta t}$) to model transition onset and progression. This model has been successfully applied to a wide range of aerodynamic configurations, including airfoils and turbomachinery blades [2,4]. Building on this foundation, subsequent advancements, such as the γ -SST model [5] and the $K\gamma$ -SST model [6], further simplified the formulation by reducing the number of transport equations or replacing them with algebraic expressions. These models have demonstrated improved accuracy in predicting transition points and flow separation in attached flows. However, their performance in massively separated flows remains limited, as the inherent averaging process of RANS formulations tends to produce predominantly two-dimensional flow structures, failing to capture the complex three-dimensional vortices characteristic of separated regions.

In parallel, hybrid RANS/LES methods have emerged as a promising approach to resolve separated flows with greater fidelity. Methods such as Detached Eddy Simulation (DES) and Improved Delayed Detached Eddy Simulation (IDDES) [7–9] combine the efficiency of RANS for boundary layer resolution with the accuracy of LES in separated regions. These methods have shown significant potential in capturing complex turbulent structures, particularly in massively separated flows. However, most existing hybrid methods are coupled with fully turbulent RANS models, which neglect the transition process and thus fail to accurately predict the boundary layer transition and its interaction with separation.

Recent efforts have sought to develop new turbulence modeling techniques that incorporate the benefits of both transitional RANS and LES approaches to simulate engineering or industrial flows more effectively. These new models aim to overcome the limitations of traditional hybrid RANS/LES methods in capturing the boundary layer transition while also reducing the high computational cost associated with LES, thereby providing a more accurate and efficient approach for predicting complex turbulent flows in practical applications. These efforts are listed in Table 1. Sorensen et al. [10] coupled the γ - $Re_{\theta t}$ -SST RANS model and the DES (detached-eddy simulation) model to predict the drag crisis of a circular cylinder. They discovered that this kind of DES- γ - $Re_{\theta t}$ model performed better than the fully turbulent DES model. Cui [11] coupled the IDDES model and the γ transition model to study the transition and separation past a wind-turbine airfoil near stall. Kwon and You [12] proposed a blending function to smoothly combine the γ - $Re_{\theta t}$ -SST RANS model with the SAS (scale adaptive simulation) or DES model to simulate the supercritical flow past a circular cylinder. Qiao et al. [13] applied the γ - $Re_{\theta t}$ -DDES (delayed detached-eddy simulation) model to simulate the attached flow past a flat plane. Yi et al. [14] adopted a γ - $Re_{\theta t}$ -IDDES (Improved Delayed Detached Eddy Simulation) model to predict the roughness-induced transition at hypersonic speed. Xiao et al. [15] coupled the k - ω - γ RANS model with the DDES model to simulate the transition and massively separated flow past the Orion capsule at hypersonic speed. Fu and Wang et al. [16] applied the DES- k - ω - γ model to predict the flow in turbomachinery. Sa [17] combined the DDES model and the γ - $Re_{\theta t}$ transition model to simulate the low-Reynolds-number flow around an Eppler 387 wing. Lin Zhou et al. [18] also applied the DDES- γ - $Re_{\theta t}$ coupling model to the sep-

arated transitional flow over the A-airfoil, DBLN-526 airfoil, and a circular cylinder in subcritical and critical regimes.

Table 1. List of hybrid RANS/LES methods coupled with transitional RANS models.

Contributors	Used Transition RANS Model	Used Hybrid RANS/LES Model	Developed Models	Test Cases
Sorensen et al. [10]	γ -Re $_{\theta t}$ -SST	DES	DES- γ -Re $_{\theta t}$	Cylinder
Wenyao et al. [11]	k- ω - γ	IDDES	IDDES-Tr	A-airfoil
Kwon and You [12]	γ -Re $_{\theta t}$ -SST	SAS	SAS1/SAS2	Cylinder
Qiao et al. [13]	γ -Re $_{\theta t}$ -SST	DDES	γ -Re $_{\theta t}$ -DDES	HGR-01 airfoil
Yi et al. [14]	γ -Re $_{\theta t}$ -SST	IDDES	γ -Re $_{\theta t}$ -IDDES	Cylinder
Xiao et al. [15]	k- ω - γ	DDES	DDES-Tr	Orion capsule
Fu and wang et al. [16]	k- ω - γ	DES	DES-k- ω - γ	Cylinder
Sa [17]	γ -Re $_{\theta t}$ -SST	DDES	DDES-Tran	E387 airfoil
Lin Zhou et al. [18]	γ -Re $_{\theta t}$ -SST	DDES	DDES- γ -Re $_{\theta t}$	A-airfoil, DBLN-526 airfoil, Cylinder
This paper	kOmegaSSTLM (γ -Re $_{\theta t}$ -SST), γ -k ω SST, K $_{\gamma}$ -SST	IDDES	γ -k ω SST-IDDES	E387, DBLN-526, NACA0012 airfoils,

OpenFOAM-v2312 is an open-source CFD software package. Currently, it includes only the γ -Re $_{\theta t}$ -SST transitional model. In this study, we first implement the γ -SST and K $_{\gamma}$ -SST transitional RANS models and then evaluate all three transitional RANS models on the E387 airfoil to identify the most accurate and computationally efficient model for predicting the boundary layer transition. Next, we develop a transitional IDDES method, referred to as γ -SST-IDDES, by coupling the γ -SST transitional RANS model with IDDES. This hybrid framework is designed to simultaneously capture the boundary layer transition and resolve massively separated flows, addressing a critical limitation of existing methods. The proposed model is rigorously validated across three airfoil cases with distinct separation characteristics: the E387 airfoil (small separation), the DBLN-526 airfoil (moderate separation), and the NACA0021 airfoil (massive separation). These test cases are selected to assess the model's capability to handle a wide range of flow conditions, from mild separations near the leading edge to massive separations at high angles of attack.

2. Numerical Methodology

2.1. Baseline Governing Equations

The k- ω Shear Stress Transport (SST) turbulence equations are used as the baseline governing equations [19,20]. The SST turbulence model has been employed to calculate the turbulent viscosity (μ_t) within the momentum and energy equations, to model the turbulence effects on the flow. The equations for the turbulent kinetic energy (k) and the specific dissipation rate (ω) are defined by Equations (1) and (2).

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = P_{k,RANS} - D_{k,RANS} + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right] \quad (1)$$

$$\frac{\partial(\rho\omega)}{\partial t} + \frac{\partial(\rho u_j \omega)}{\partial x_j} = \frac{\alpha\omega}{k} P_k - D_\omega + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_\omega \mu_t) \frac{\partial \omega}{\partial x_j} \right] + \frac{2(1 - F_1)(\rho \sigma_\omega \omega^2)}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \quad (2)$$

where ρ is the density, μ is the dynamic viscosity, and u_j is the flow velocity. The production and destruction terms for the k equation are defined by Equations (3) and (4), respectively.

$$P_{k,RANS} = \min(P_k, 10 * \beta^* \rho \omega k); \quad P_k = \mu_t \frac{\partial u_j}{\partial x_i} \left[\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right] \quad (3)$$

$$D_{k,RANS} = \frac{\rho k^{\frac{3}{2}}}{l_{RANS}}; \quad l_{RANS} = \frac{k^{\frac{1}{2}}}{\omega \beta^*} \quad (4)$$

2.2. Transitional RANS Models

2.2.1. kOmegaSSTLM (γ -Re $_{\theta t}$ -SST)

The kOmegaSSTLM (γ -Re $_{\theta t}$ -SST) model proposed by Menter et al. [1–4] contains two new transport equations—the momentum-thickness Reynolds number ($Re_{\theta t}$) and intermittency (γ), represented by Equations (5) and (6), respectively.

$$\frac{\partial(\rho \tilde{Re}_{\theta t})}{\partial t} + \frac{\partial(\rho U_j \tilde{Re}_{\theta t})}{\partial x_j} = P_{\theta t} + \frac{\partial}{\partial x_j} \left[\sigma_{\theta t} \left(\mu + \frac{\mu_t}{\mu_t} \right) \frac{\partial \tilde{Re}_{\theta t}}{\partial x_j} \right] \quad (5)$$

$$\frac{\partial(\rho \gamma)}{\partial t} + \frac{\partial(\rho U_j \gamma)}{\partial x_j} = P_\gamma - E_\gamma + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\gamma} \right) \frac{\partial \gamma}{\partial x_j} \right] \quad (6)$$

where E_γ and P_γ are the destruction and transition source defined by Equations (7) and (8), respectively.

$$E_\gamma = c_{a2} \rho \Omega \gamma F_{turb} (C_{e1} \gamma - 1) \quad (7)$$

$$P_\gamma = F_{length} c_{a1} \rho S [\gamma F_{onset}]^{0.5} (1 - C_{e1} \gamma) \quad (8)$$

where F_{length} and F_{onset} are dimensionless functions that are used to control the intermittency equation within the boundary layer, and both of these functions are dependent on $\tilde{Re}_{\theta t}$.

Finally, γ -Re $_{\theta t}$ equations interact with the kinetic energy equation (Equation (1)) of the SST model, resulting in a modified kinetic energy equation for the γ -Re $_{\theta t}$ -SST model, which takes the form of Equation (9).

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = \tilde{P}_k - \tilde{D}_k + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right] \quad (9)$$

$$\tilde{P}_k = \gamma_{eff} P_{k,RANS}; \quad \tilde{D}_k = \min(\max(\gamma_{eff}, 0.1), 1.0) D_{k,RANS}$$

where $P_{k,RANS}$ and $D_{k,RANS}$ are the original production and destruction terms for the $k\omega$ SST model in Equations (3) and (4), respectively.

2.2.2. γ -SST

The γ -SST transition model was proposed by Menter et al. [5]. This transitional model consists of only one transport equation for the intermittency (γ), Equation (10).

$$\frac{\partial(\rho \gamma)}{\partial t} + \frac{\partial(\rho U_j \gamma)}{\partial x_j} = P_\gamma - E_\gamma + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\gamma} \right) \frac{\partial \gamma}{\partial x_j} \right] \quad (10)$$

where the transition source term P_γ is defined as

$$P_\gamma = F_{length} \rho S \gamma (1 - \gamma) F_{onset} \quad (11)$$

The magnitude of this source term is controlled by the transition length function, F_{length} , which used to be a correlation [4], but has been changed to a constant value of 100 [5]. The formulation of the function F_{onset} is used to trigger intermittency production. It contains the ratio of the local vorticity Reynolds number (Re_v) to the critical Reynolds number ($Re_{\theta C}$). The transition onset is controlled by the following functions:

$$\begin{aligned} F_{onset1} &= \frac{Re_v}{2.2 Re_{\theta C}} \\ F_{onset2} &= \min(F_{onset1}, 2.0) \\ F_{onset3} &= \max\left(1 - \left(\frac{R_T}{3.5}\right)^3, 0\right) \\ F_{onset} &= \max(F_{onset2} - F_{onset3}, 0) \end{aligned} \quad (12)$$

The γ -SST model is finally obtained by coupling the γ equation with the SST turbulence. The new turbulence kinetic energy k equation of γ -SST is defined by Equation (13).

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = \tilde{P}_k + P_k^{lim} - \tilde{D}_{k,\gamma,RANS} + \frac{\partial}{\partial x_j} \left((\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right) \quad (13)$$

$$\begin{aligned} \tilde{P}_k &= \gamma P_{k,RANS} \\ P_k^{lim} &= 5C_k \max(\gamma - 0.2, 0) (1 - \gamma) F_{on}^{lim} \max(3C_{SEP} \mu - \mu_t, 0) S \Omega \\ F_{on}^{lim} &= \min\left(\max\left(\frac{Re_v}{2.2 Re_{\theta C}^{lim}} - 1, 0\right), 3\right); \quad Re_{\theta C}^{lim} = 1100 \\ \tilde{D}_{k,\gamma,RANS} &= \max(\gamma, 0.1) D_{k,RANS} \end{aligned} \quad (14)$$

where $P_{k,RANS}$ and $D_{k,RANS}$ are the original production and destruction terms for the $k\omega$ SST model in Equations (3) and (4), respectively.

2.2.3. $K\gamma$ -SST

The $K\gamma$ transition model was proposed by Sandhu et al. [6]. The $K\gamma$ - ω SST transitional model is derived from γ - $k\omega$ SST; however, it uses a simple algebraic term, as shown in Equation (15), to estimate the intermittency using the turbulent Reynolds number (R_T) instead of transport equations.

$$\gamma \approx \tilde{\gamma} = \left(1 - e^{-R_T}\right)^n, \text{ where } n = 3, R_T = \frac{\rho k}{\omega \mu} \quad (15)$$

Furthermore, the $K\gamma$ transition model interacts with the turbulent kinetic energy equation (Equation (1)), leading to the modification of the kinetic energy equation into the following form:

$$\frac{\partial(\rho K_\gamma)}{\partial t} + \frac{\partial(U_j \rho K_\gamma)}{\partial x_j} = P_{K_\gamma}^{(k)} + P_{K_\gamma}^{lim} - E_{K_\gamma}^{(k)} + D_{K_\gamma}^{(k)} + P_{K_\gamma}^{(\gamma)} - E_{K_\gamma}^{(\gamma)} + D_{K_\gamma}^{(\gamma)} \quad (16)$$

where $P^\varnothing$, $E^\varnothing$, $D^\varnothing$ represent the production, destruction, and diffusion terms in the transport equation of the variable $\varnothing \in \{k, \gamma\}$, respectively. The source terms are defined as follows:

$$\begin{aligned} P_{K_\gamma}^{(k)} &= \tilde{\gamma} \mu_t S \Omega \\ P_{K_\gamma}^{(\gamma)} &= F_{length} \rho S (1 - \tilde{\gamma}) F_{onset} k \\ E_{K_\gamma}^{(k)} &= \max(\tilde{\gamma}, 0.1) \beta^* \rho \omega k; \\ E_{K_\gamma}^{(\gamma)} &= \tilde{\gamma} C_{e1} \rho \Omega F_{turb} k \end{aligned} \quad (17)$$

More comprehensive information regarding this transitional model can be found in reference [6].

2.3. Improved Delayed Detached Eddy Simulation (IDDES) Formulations

IDDES is a type of DES-hybrid RANS/LES method [21,22], which possesses the capability to adaptively employ either DDES [23] or WMLES (Wall-Modeled LES) functionalities in different flow regions. In areas dominated by vortices, DDES length scales are implemented to eliminate grid scale dependence in the original DES methods, and in the near-wall regions, the WMLES model is employed to accurately capture boundary-layer profiles and eliminate the log-layer mismatch problem commonly encountered in DES hybrid methods. Consequently, the IDDES model combines the strengths of both DDES and WMLES techniques, resulting in improved predictions of small separation flows compared to the previous methods.

In the construction of RANS/LES hybrid methods, the grid scale (Δ) in the LES methods is introduced into turbulence models. The LES-type grid scale is given by

$$\begin{aligned} l_{LES} &= C_{DES} \Delta \\ \Delta &= \min\{\max[C_w d_w, C_w h_{max}, h_{wn}], h_{max}\} \end{aligned} \quad (18)$$

where C_w is taken as 0.15; d_w is the wall distance; h_{max} is the maximum cell length in three directions, which is the same definition as the original DES; and h_{wn} is the normal cell height near the surface walls. The DDES length scale is formulated as hybrid RANS and LES length scales, incorporating a flow-dependent blending function (f_d) which effectively prevents the LES mode from switching early in locally refined boundary-layer meshes [24].

$$l_{DDES} = l_{RANS} - f_d \max\{l_{RANS} - l_{LES}\} \quad (19)$$

where the blending function (f_d) is smooth in the transition area from the RANS to LES mode and given by

$$f_d = 1 - \tanh\left[(8r_d)^3\right] \quad (20)$$

where

$$r_d = \frac{v + \nu_t}{k^2 d_w^2 \cdot \max\left\{\left[\sum_{ij} (\partial u_i / \partial x_j)^2\right]^{1/2}, 10^{-10}\right\}} \quad (21)$$

The WMLES length scale is defined as

$$l_{WMLES} = f_B (1 + f_e) l_{RANS} + (1 - f_B) l_{LES} \quad (22)$$

where f_B is another blending function defined as

$$\begin{aligned} f_B &= \min\{2\exp(-9\alpha^2), 1.0\} \\ \alpha &= 0.25 - \frac{d_w}{h_{max}}, \\ f_e &= \max[(f_{e1} - 1), 0] f_{e2} \end{aligned} \quad (23)$$

The IDDES length scale is a combination of DDES and WMLES length scales, which is given as

$$\begin{aligned} l_{IDDES} &= \tilde{f}_d \cdot (1 + f_e) \cdot l_{k-\omega} + (1 - \tilde{f}_d) \cdot l_{LES} \\ \tilde{f}_d &= \max\{(1 - f_{dt}), f_B\} \end{aligned} \quad (24)$$

2.3.1. $k\omega$ SST-IDDES Formulation

The $k\omega$ SST-IDDES model combines the fully turbulent RANS model $k\omega$ SST with the IDDES model. In $k\omega$ SST-IDDES, the length scale of RANS (l_{RANS}) in the destruction term of the turbulent kinetic energy equation (Equation (1)) is eventually replaced with the length scale of IDDES (l_{IDDES}) given in Equation (24).

$$\tilde{D}_{k,IDDES} = \frac{\rho k^{\frac{3}{2}}}{l_{IDDES}} \quad (25)$$

Hence, the k equation of $k\omega$ SST-IDDES is represented by Equation (26):

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = P_{k,RANS} - \tilde{D}_{k,IDDES} + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right] \quad (26)$$

where $P_{k,RANS}$ is the original production term for the $k\omega$ SST model in Equation (3).

2.3.2. γ - $k\omega$ SST-IDDES Formulation

The γ - $k\omega$ SST-IDDES model couples the transitional RANS γ - $k\omega$ SST with the IDDES model. It is ultimately derived by substituting the l_{RANS} of the destruction term in the kinetic energy equation (Equation (13)) of the γ - $k\omega$ SST model with the l_{IDDES} represented by Equation (22). Equation (27) represents the new kinetic energy equation of the γ - $k\omega$ SST-IDDES model, while the ω equation remains unchanged.

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = \tilde{P}_k + P_k^{lim} - \max(\gamma, 0.1) \frac{\rho k^{\frac{3}{2}}}{l_{IDDES}} + \frac{\partial}{\partial x_j} \left((\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right) \quad (27)$$

3. Numerical Results and Discussion

3.1. Two-Dimensional Simulations Around E387 Airfoil

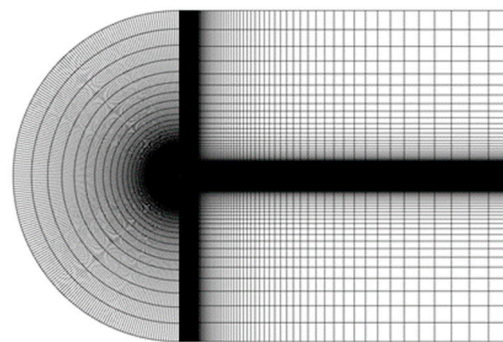
The simulations are performed on the two-dimensional E387 airfoil test-case in accordance with the published wind tunnel test by McGhee et al. [25]. The details of the experimental flow parameters are presented in Table 2, which are strictly satisfied in the numerical simulations.

Figure 1 depicts the “CO-grid” mesh topology that is used in the computational area. The first layer spacing is set to 2.0×10^{-5} to make sure $y^+ < 1$. A small O-grid is generated with growing boundary layers around the E387 airfoil, as illustrated in Figure 1a. The diameter of the O-grid is three times that of the chord (c) of the airfoil. The expansion ratio from the first layer to the O-grid boundary is set to 1.2 and is constant for both the leading and trailing edges. The C-grid expansion begins from the last cell size of the O-grid, continuing with the same ratio. The C-grid is extended to $20c$ in the upstream direction

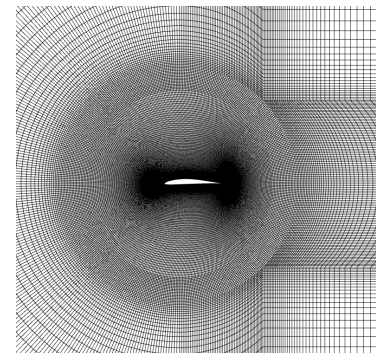
and 40c in the downstream direction and is located outside of the O-grid. Since all the simulations are two-dimensional, the CO-grid is extruded with one chord (c) length by one spanwise layer in the spanwise direction.

Table 2. Flow parameters and working conditions.

Parameter	Notation	Value
Reynolds number	Re	300,000
Chord length	c	1 m
Angle of attack	α	$0^\circ, 2^\circ, 4^\circ, 6^\circ$
Turbulence Intensity	Tu	0.1%
Turbulent/Laminar kinematic viscosity	ν_t/ν	10



(a) CO-grid on E387 two-dimensional test case



(b) Zoomed in O-grid around

Figure 1. CO-grid on E387 two-dimensional test case: (a) CO-grid around airfoil; (b) Zoomed in O-grid around.

A pressure-based, incompressible steady flow solver, simpleFOAM, is used to perform the two-dimensional simulations. It employs the well-established SIMPLE algorithm [26] within the finite volume framework to solve the steady-state incompressible Navier–Stokes equations, with pressure–velocity coupling handled by the SIMPLE procedure. A steady-state approach is used for the time derivative term. Spatial discretization employs the 2nd cell-limited Gauss linear scheme for all gradient calculations in order to maintain numerical stability. For the divergence terms, a bounded Gauss linear upwind scheme based on velocity gradients is applied to the momentum equations, while viscous stress terms use the Gauss linear scheme. Different boundary conditions are assigned for different patches and linear solvers as represented by Tables 3 and 4, respectively.

Table 3. Boundary conditions.

Variable	Airfoil	Inlet	Outlet	FrontAndBack
p	zeroGradient	freestreamPressure	freestreamPressure	empty
U	noSlip	freestreamVelocity	freestreamVelocity	empty
k	fixedValue	fixedValue	zeroGradient	empty
omega	fixedValue	fixedValue	zeroGradient	empty
nut	nutLowReWallFunction	freestream	Freestream	empty
gammaInt	zeroGradient	fixedValue	zeroGradient	empty
ReThetat	zeroGradient	fixedValue	zeroGradient	empty

Four meshes, designated as mesh AA, mesh BB, mesh CC, and mesh DD, have been created for the purpose of conducting a grid convergence study, as outlined in Table 5. There are 110, 288, 344, and 544 wrap-around points on the surface of the airfoil, respectively. Figure 2 shows that the pressure coefficient (C_p) distributions on the airfoil's surface are

presented at an angle of attack of 0° with the γ - $k\omega$ SST transitional model. It is observed that the C_p curve exhibits good convergence with the experimental data, and mesh CC and mesh DD demonstrate a high degree of similarity. Consequently, mesh CC has been selected for the simulations of other angle of attacks.

Table 4. Linear Solvers.

Variable	Solver	Tolerance	Relative Tolerance
p	GAMG	1×10^{-8}	0.001
omega	PBiCG	1×10^{-8}	0
U/k/gammaInt/Rethetat	PBiCG	1×10^{-14}	0.0001

Table 5. Details of the grid.

Grid	Wrap-Around Points	Points Normal to Surface
Mesh AA	110	99
Mesh BB	288	129
Mesh CC	344	163
Mesh DD	544	200

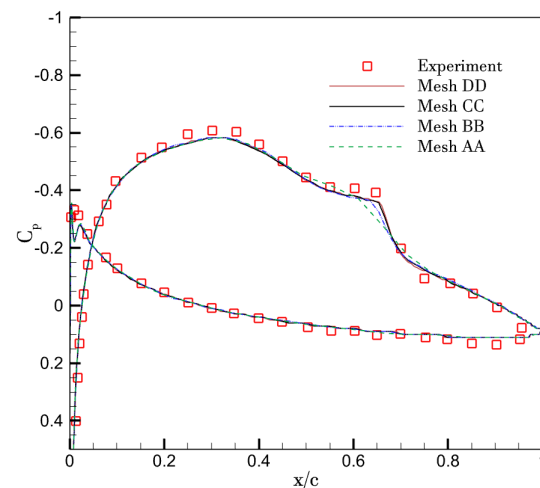
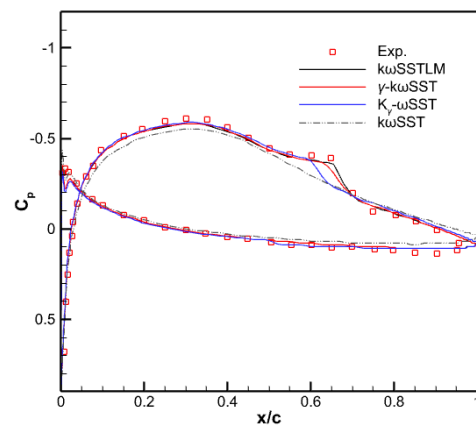


Figure 2. Grid convergence study for E387 airfoil at 0° .

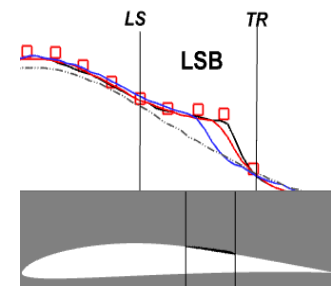
Figure 3 presents the pressure coefficient (C_p) curve of the E387 airfoil at $\alpha = 2^\circ$. Figure 4 depicts the streamlines near the laminar separation bubble (LSB). The positions of the Laminar Separation Point (LS) and Turbulent Reattachment Point (TR) are determined and the corresponding results are presented in Table 6. It is observed that the $k\omega$ SSTLM model offers the most accurate estimation of the transition point within the LSB followed by γ - $k\omega$ SST and $K\gamma$ - ω SST, respectively.

Table 6. Details of the LSB at $\alpha = 2^\circ$.

Transition Model	LS-TR Location (x/c)	LSB	Transition Point (x/c)
$k\omega$ SSTLM	0.518–0.676	0.158	0.646
γ - $k\omega$ SST	0.529–0.659	0.130	0.629
$K\gamma$ - ω SST	0.538–0.604	0.066	0.592
Experiment	0.5–0.7	0.2	0.660



(a) Pressure Coefficients



(b) Zoomed-In pressure coefficients near LSB

Figure 3. Coefficient of pressure of an E387 airfoil at $\alpha = 2^\circ$: (a) Pressure Coefficients; (b) Zoomed-In pressure coefficients near LSB.

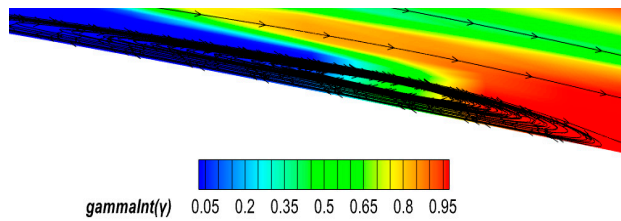
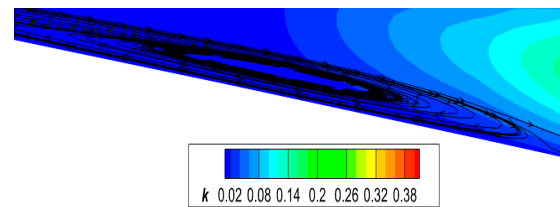
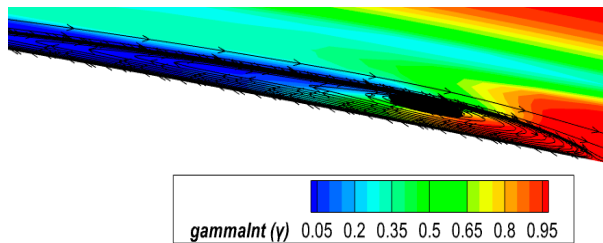
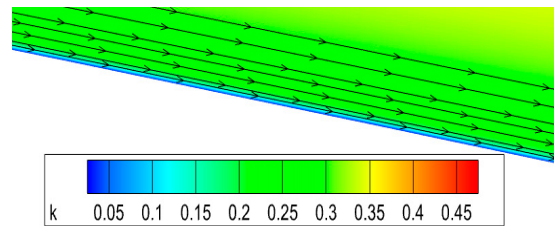
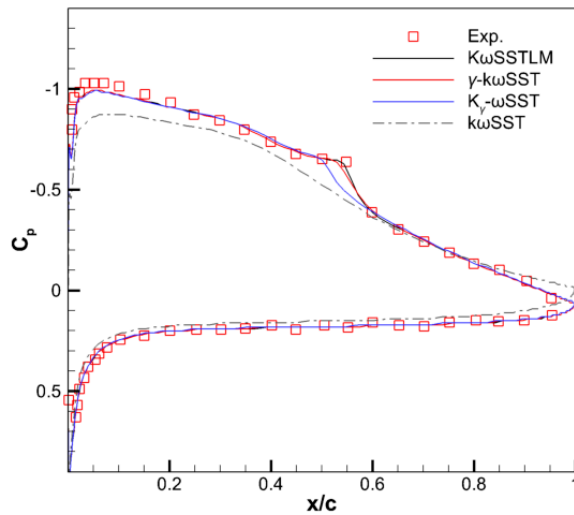
(a) γ -k ω SST(b) K γ - ω SST(c) k ω SSTLM(d) k ω SST (No LSB)

Figure 4. LSB on upper surface of E387 airfoil at $\alpha = 2^\circ$: (a) γ -k ω SST; (b) K γ - ω SST; (c) k ω SSTLM; (d) k ω SST (No LSB).

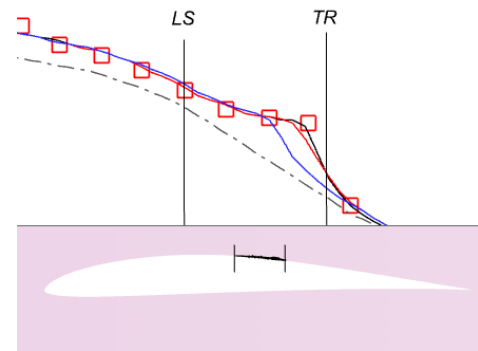
Figure 5 illustrates the C_p curve for the E387 airfoil at $\alpha = 4^\circ$. Table 7 summarizes the computational result of the LSB span and transition location estimated by each transitional model, as represented by Figure 6. When comparing the LS point at $\alpha = 4^\circ$ with that at $\alpha = 2^\circ$, an apparent forward shift in the LSB, towards the leading edge, is observed. The k ω SSTLM model estimates an LSB shift of $0.064 x/c$, the γ -k ω SST model predicts a shift of $0.077 x/c$, and the K γ - ω SST model anticipates a shift of $0.062 x/c$.

Table 7. Details of the LSB at $\alpha = 4^\circ$.

Transition Model	LS-TR Location (x/c)	LSB Size	Transition Point (x/c)
k ω SSTLM	0.454–0.561	0.107	0.542
γ -k ω SST	0.452–0.553	0.101	0.524
K γ - ω SST	0.476–0.501	0.025	0.490
Experiment	0.4–0.575	0.175	0.550

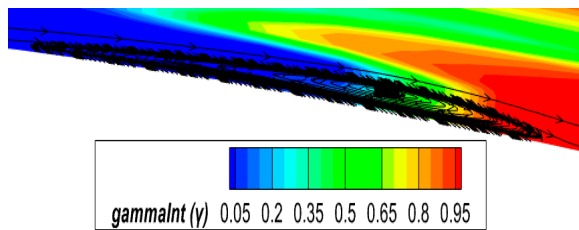


(a) Pressure Coefficients

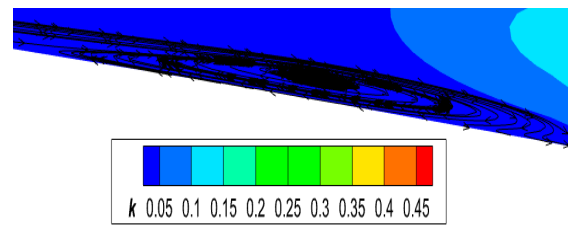


(b) Zoomed-In pressure coefficients near LSB

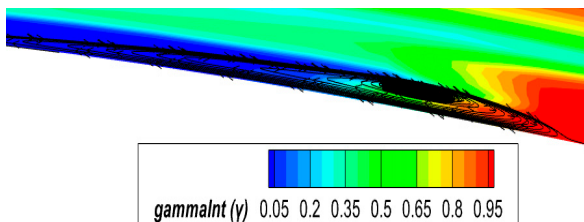
Figure 5. Coefficient of pressure of an E387 airfoil at $\alpha = 4^\circ$: (a) Pressure Coefficients; (b) Zoomed-In pressure coefficients near LSB.



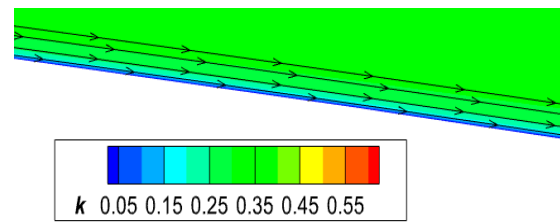
(a) γ - $k\omega$ SST



(b) $K\gamma$ - ω SST



(c) $k\omega$ SSTLM



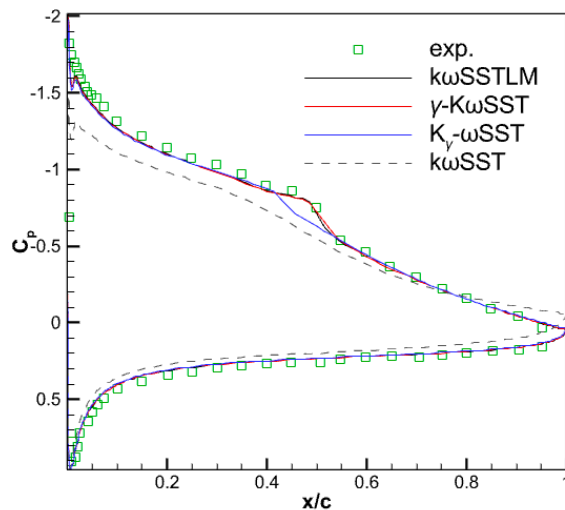
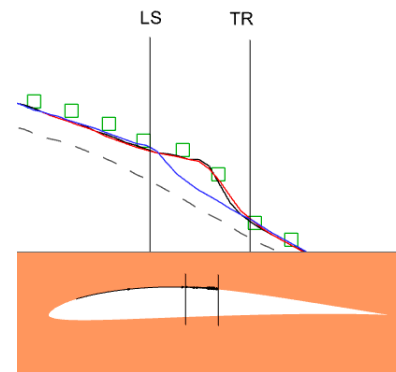
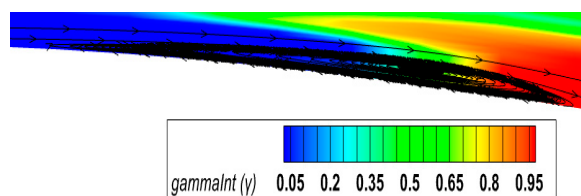
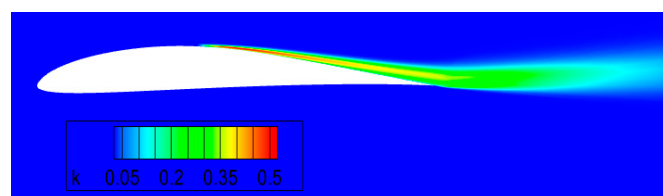
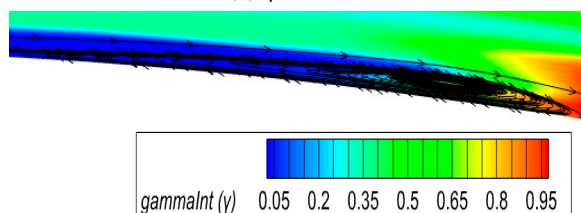
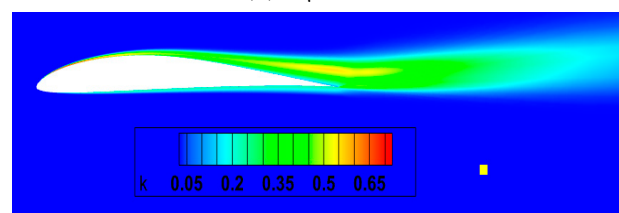
(d) $k\omega$ SST (No LSB)

Figure 6. LSB on upper surface of E387 airfoil at $\alpha = 4^\circ$: (a) γ - $k\omega$ SST; (b) $K\gamma$ - ω SST; (c) $k\omega$ SSTLM; (d) $k\omega$ SST (No LSB).

Figure 7 illustrates the C_p curve for the E387 airfoil at $\alpha = 6^\circ$. At this particular α , the $K\gamma$ - ω SST model fails to capture the LSB streamlines, as they entirely vanish. However, the γ - $k\omega$ SST and $k\omega$ SSTLM models successfully captured the LSB streamlines, as represented in Figure 8. Table 8 summarizes the computational results of the LSB span and transition location. Upon comparing the transition behavior between $\alpha = 2^\circ$ and $\alpha = 6^\circ$, it is observed that the LSB moves even closer towards the leading edge. The $k\omega$ SSTLM model estimates a forward LSB shift of $0.12 x/c$, while the γ - $k\omega$ SST model predicts a shift of $0.133 x/c$.

Table 8. Details of the LSB at $\alpha = 6^\circ$.

Transition Model	LS-TR Location (x/c)	LSB Size	Transition Point (x/c)
$k\omega$ SSTLM	0.398–0.498	0.100	0.479 x/c
γ - $k\omega$ SST	0.396–0.500	0.104	0.474 x/c
$K\gamma$ - ω SST	Absence of LSB streamlines	-	0.470 x/c
Experiment	0.38–0.46	0.08	0.46 x/c

**(a)** Pressure Coefficients**(b)** Zoomed-In pressure coefficients near LSB**Figure 7.** Coefficient of pressure of an E387 airfoil at $\alpha = 6^\circ$: **(a)** Pressure Coefficients; **(b)** Zoomed-In pressure coefficients near LSB.**(a)** γ - $k\omega$ SST**(b)** $K\gamma$ - ω SST**(c)** $k\omega$ SSTLM**(d)** $k\omega$ SST (No LSB)**Figure 8.** LSB on upper surface of E387 airfoil at $\alpha = 6^\circ$: **(a)** γ - $k\omega$ SST; **(b)** $K\gamma$ - ω SST; **(c)** $k\omega$ SSTLM; **(d)** $k\omega$ SST (No LSB).

The results from this 2D test case show that the LSB shifts forward towards the leading edge when the angle of attack is increased. Furthermore, the $k\omega$ SSTLM and γ - $k\omega$ SST transitional models produced comparable results with a slight deviation. Therefore, in this paper, the γ - $k\omega$ SST transitional model is employed to couple with IDDES.

3.2. IDDES and URANS Simulations

All the IDDES and URANS simulations are carried out using the rhoPimpleFoam solver within the OpenFOAM v2312. This solver implements the PIMPLE algorithm, which is a combination of the SIMPLE [27] and PISO [28] algorithms. This approach computes a transient and compressible solution, offering greater stability than the PISO algorithm alone and allowing for the use of larger time steps. The rhoPimpleFoam solver employs the finite volume method for numerical representation of the compressible three-dimensional Navier–Stokes equations. As suggested by Robertson [29], a standard three-level second-order backward difference is used for the time marching scheme. For spatial discretization, the 2nd-order cell-based scalar-limiting central difference is used for the gradient term; the 2nd-order bounded central difference is used for the divergence scheme; and the 2nd-order limited deferred correction scheme is used for the Laplacian scheme. In order to speed up the calculations, the message passing interface (MPI) is used for parallel computing.

The numerical test cases involve transitional flow over the E387 airfoil, DBLN-526 airfoil, and NACA0021 airfoil. These cases include a range of separation behaviors, small separation for the E387 airfoil, moderate separation for the DBLN-526 airfoil, and massive separation for the NACA0021 airfoil, to evaluate the performance of γ -k ω SST-IDDES in separated transitional flows.

3.2.1. Flow over an E387 Airfoil

Jeong Hwan Sa et al. [17] used the E387 airfoil to verify their $\gamma\overline{R_{\theta t}}$ -DDES model. Their observations revealed that both the transitional IDDES model and the fully turbulent IDDES model generated the LSB at the leading edge because of the small separation. The transitional IDDES model produced an LSB size ranging from $x/c = 0.025$ to $x/c = 0.091$, while the fully turbulent IDDES model produced an LSB ranging from $x/c = 0.004$ to $x/c = 0.043$.

The numerical simulations are performed to simulate the flow over the E387 airfoil at an angle of attack of 11° and a Reynolds number of 1×10^5 . The freestream turbulence intensity is specified as $Tu_\infty = 0.1\%$. The “CO-grid” mesh topology is adopted in the computational domain as shown in Figure 9. A small O-grid is generated with 524 points in the circumferential direction, 311 points on the suction side and 213 points on the pressure side, along with 101 points in the normal direction, with the diameter of this O-grid being $3c$. The first cell normal to the wall is placed at $1 \times 10^{-5}c$, so that $y^+ < 0.8$. The C-grid is positioned on the outside of the O-grid and extends to $25c$ in the upstream direction and $40c$ in the downstream direction, with 51 and 101 grid points in the respective directions. The spanwise width is set to $z/c = 0.2$, and three sets of grids with $N_z = 25$, $N_z = 35$, and $N_z = 49$ are generated to study the effect of spanwise grid resolution. A non-dimensional time step size $\Delta t U_\infty C^{-1} = 0.001$ is used for all unsteady time-accurate computations. A total of 15 non-dimensional time units are computed, and the time-averaged solutions were obtained by taking the average over the last seven non-dimensional time units. The boundary conditions and the linear solver settings in OpenFOAM-v2312 for IDDES and URANS simulations in this paper are detailed in Tables 9 and 10, respectively.

Table 9. Boundary conditions for IDDES and URANS simulations.

Variable	Airfoil	Inlet	Outlet	Front/Back
p	zeroGradient	zeroGradient	fixedValue	symmetryPlane
U	noSlip	fixedValue	zeroGradient	symmetryPlane
k	kqRWallFunction	fixedValue	zeroGradient	symmetryPlane
omega	omegaWallFunction	fixedValue	zeroGradient	symmetryPlane
nut	nutUSpaldingWallFunction	zeroGradient	zeroGradient	symmetryPlane
gammaInt	zeroGradient	fixedValue	zeroGradient	symmetryPlane

Table 10. Linear Solver settings for IDDES and URANS simulations.

Variable	Solver	Preconditioner	Tolerance	Relative Tolerance
p	PCG	GAMG	1×10^{-8}	0.01
U/k/gammaInt/omega	PBiCGStab	DILU	1×10^{-8}	0.001

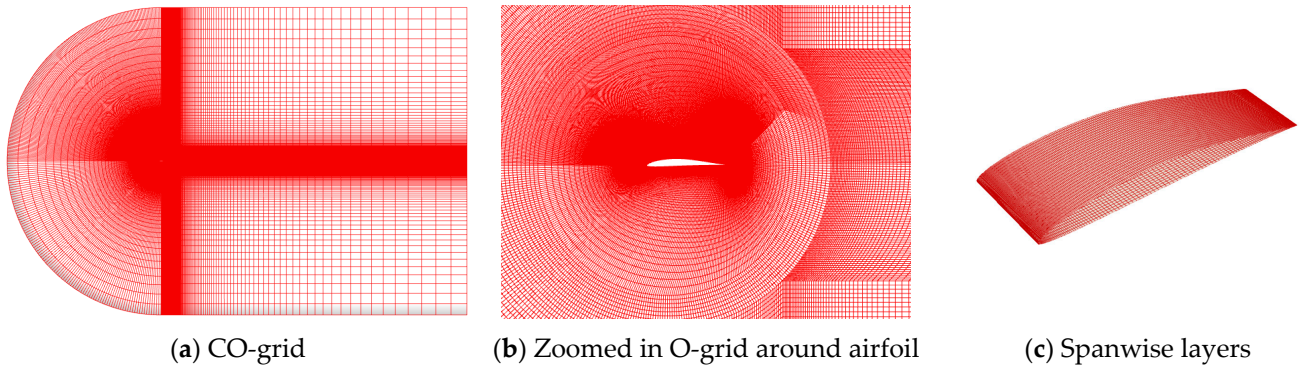
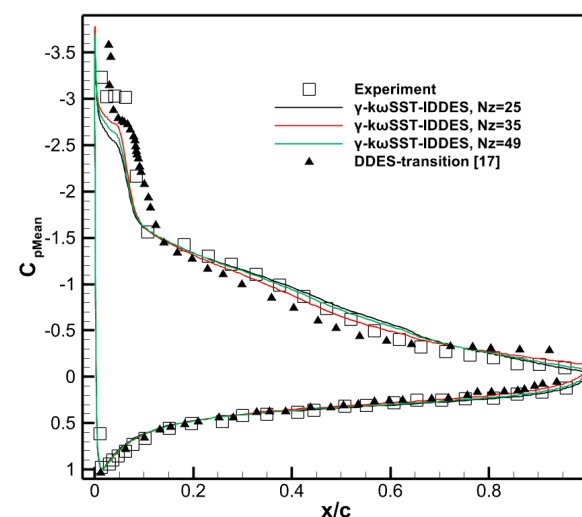
**Figure 9.** CO-grid topology around E387 airfoil: (a) CO-grid; (b) Zoomed in O-grid around airfoil; (c) Spanwise layers.

Figure 10 illustrates the distributions of the time-averaged surface pressure coefficient at three different numbers of grids. This C_p plot shows the best agreement with experimental data when $N_z = 35$. Therefore, $N_z = 35$ is selected for further simulations. Figure 11a,b depicts the time history of lift and drag coefficients, respectively, with the corresponding time-averaged force coefficients summarized in Table 11. It can be observed from this table that both simulations predict similar force coefficients, possibly due to the small angle of attack. However, the URANS: γ - $k\omega$ SST simulation yields significantly lower values compared to the IDDES.

Table 11. Comparisons of time-averaged C_L and C_D for E387 airfoil test case.

Case	C_L	C_D
URANS- γ - $k\omega$ SST	1.13633	0.03977
γ - $k\omega$ SST-IDDES	1.23946	0.04330
$k\omega$ SST-IDDES	1.24641	0.04446
Experiment [25]	1.201	0.0525
DDES-transition [17]	1.152	0.0663

**Figure 10.** Distribution of the pressure coefficient C_p of the time-averaged flow field for the E387 airfoil.

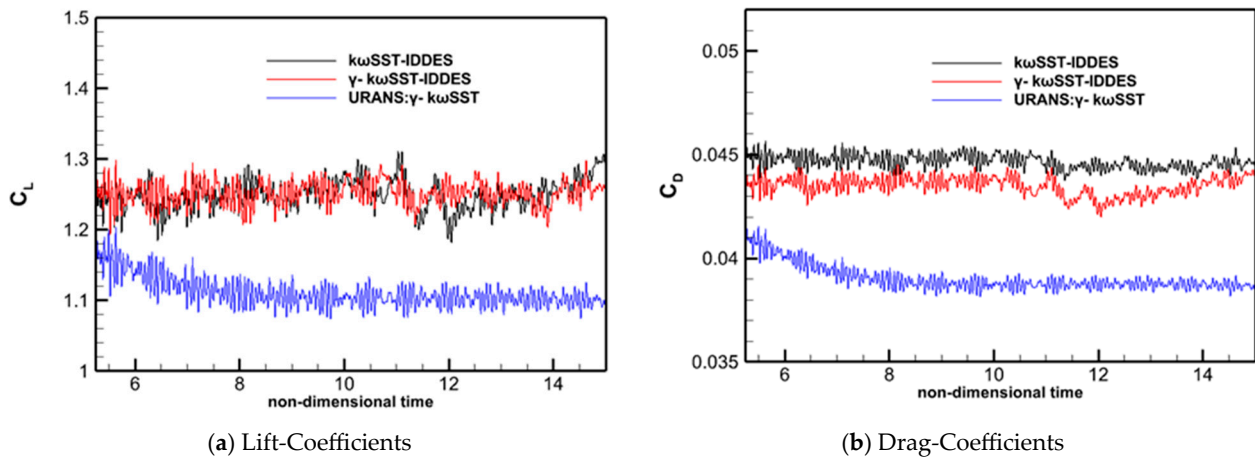


Figure 11. Time history of lift and drag coefficient for E387 airfoil: (a) Lift-Coefficients; (b) Drag-Coefficients.

Figure 12 shows the vortex structure indicated by the $Q = \frac{1}{2}[(\text{tr}(\nabla U))^2 - \text{tr}(\nabla U \cdot \nabla U)]$ distributions. The URANS model exhibits small separation close to the leading edge, whereas the IDDES models generate intricate vortex formations. These vortex patterns, characteristic of Large Eddy Simulations, demonstrate the effective integration of the IDDES and transition model within the present γ - $k\omega$ SST-IDDES model.

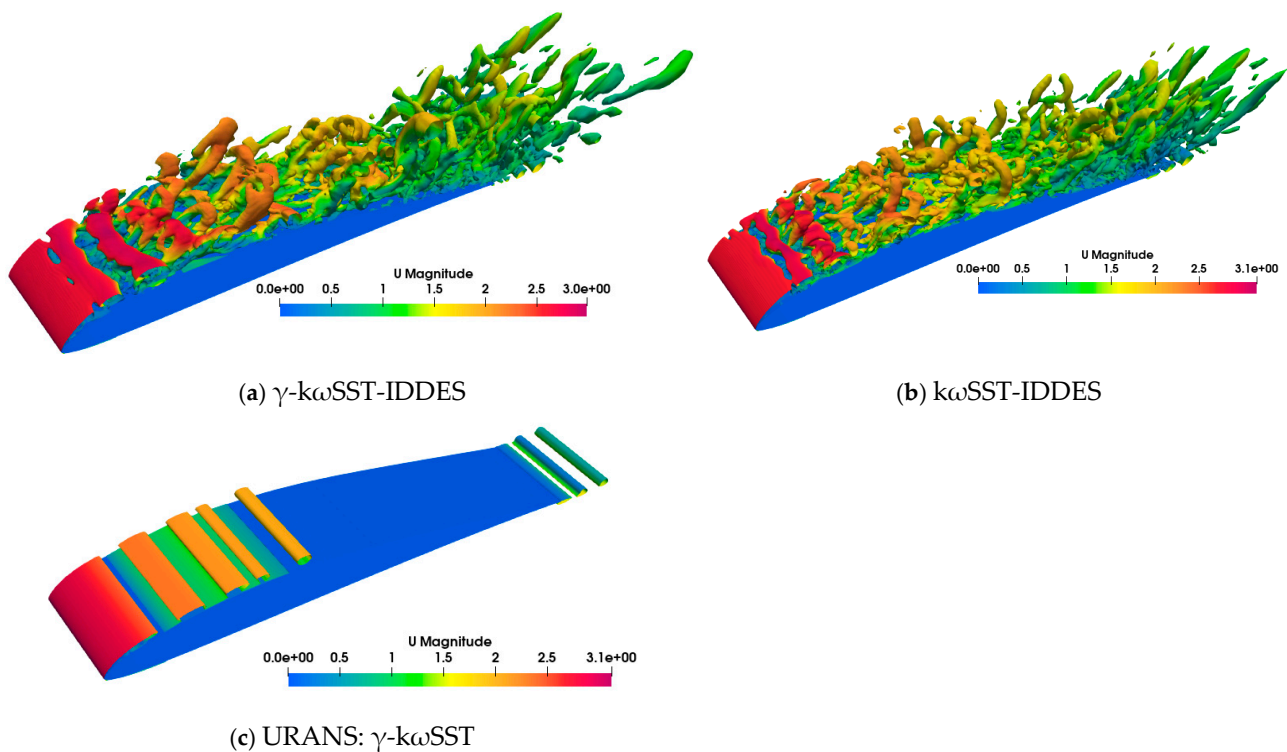


Figure 12. Instantaneous flow field results for E387 airfoil test case (Q -criterion = 100 isosurface colored by the velocity magnitude): (a) γ - $k\omega$ SST-IDDES; (b) $k\omega$ SST-IDDES; (c) URANS: γ - $k\omega$ SST.

Figure 13 shows the time-averaged streamlines near the LSB. The $k\omega$ SST-IDDES yields a considerably smaller LSB when compared to the γ - $k\omega$ SST-IDDES. Specifically, the LSB size produced by the γ - $k\omega$ SST-IDDES ranges from $x/c = 0.063$ to $x/c = 0.121$, while the LSB generated by the $k\omega$ SST-IDDES spans from $x/c = 0.041$ to $x/c = 0.077$.

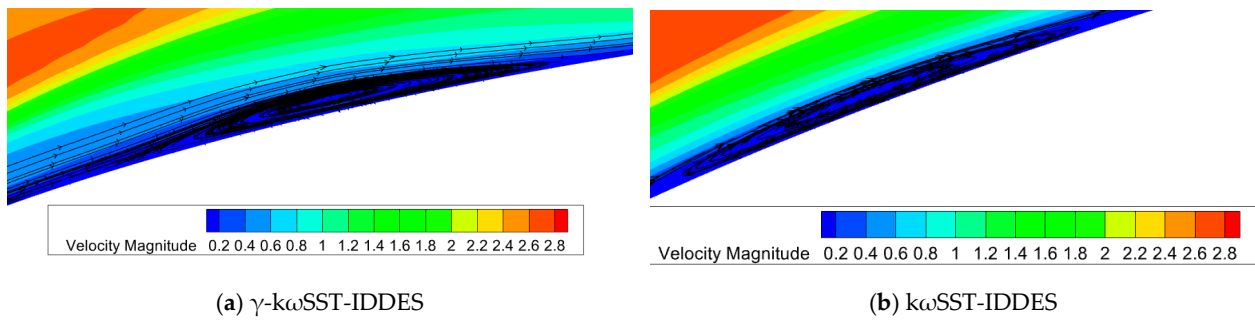


Figure 13. Time-averaged streamlines near the laminar separation bubble for E387 airfoil: (a) γ - $k\omega$ SST-IDDES; (b) $k\omega$ SST-IDDES.

3.2.2. Flow over a DBLN-526 Airfoil

The DBLN-526 airfoil was first used by Lin Zhou et al. [18] to validate their $\gamma\overline{R_{\theta t}}$ -DDES transitional model along with a wind tunnel experiment at the NLF wind tunnel of Northwestern Polytechnical University. In the experiments, the transition point was observed near the leading edge at $x/c \approx 0.05$, and the boundary layer separates around $x/c \approx 0.5$.

In this section, numerical simulations are conducted to analyze the flow over the DBLN-526 airfoil at an angle of attack of 20° with a chord-based Reynolds number of 0.8×10^6 and a turbulence intensity of 0.15%. The “CO-grid” mesh topology is also utilized for this test case in the computational domain as shown in Figure 14. A small O-grid is created with 568 points in the circumferential direction, comprising 287 points on the suction side and 281 points on the pressure side. Additionally, there are 141 points in the normal direction, and the diameter of this O-grid is set at 6 times the chord length ($6c$). The height of the first grid normal to the wall is placed at $1 \times 10^{-5}c$, ensuring $y^+ < 1$. The C-grid is situated outside the O-grid and extends to $25c$ in the upstream direction and $40c$ in the downstream direction, which consists of 81 grid points upstream and 141 grid points downstream. The spanwise width is extended to $0.5c$, and four sets of grids with $N_z = 31$, $N_z = 41$, $N_z = 45$, and $N_z = 51$ are used. A non-dimensional time step size $\Delta t U_\infty C^{-1} = 0.0018$ is used for all unsteady time-accurate computations. The computations are carried out for a total of 10 non-dimensional time units. The time-averaged solutions are obtained by averaging the data over the last six non-dimensional time units.

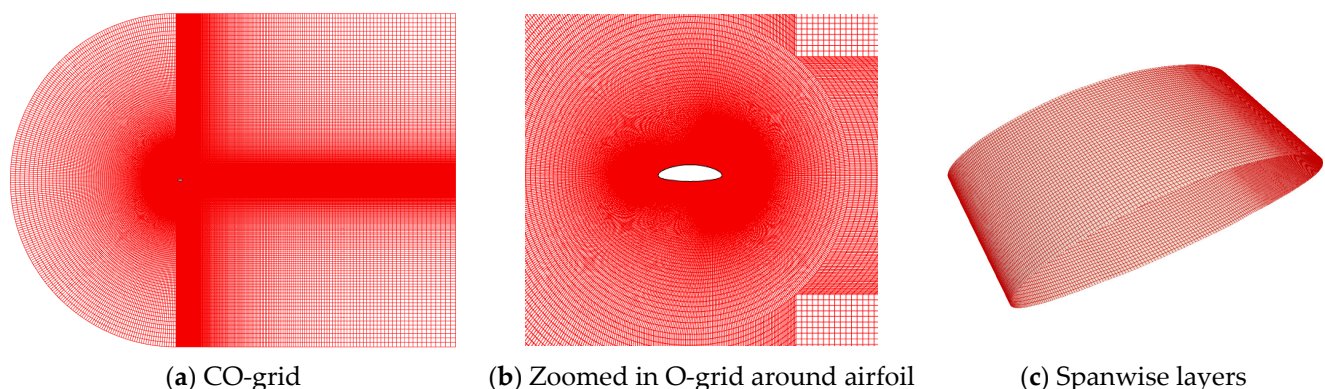


Figure 14. CO-grid topology around DBLN-526 airfoil: (a) CO-grid; (b) Zoomed in O-grid around airfoil; (c) Spanwise layers.

Figure 15 shows the time-averaged surface pressure coefficient distribution at four different levels of grids. The present method, γ - $k\omega$ SST-IDDES, shows the grid convergence beyond $N_z = 41$, and agrees well with the experiment and computation of Zhou [18].

Figure 16 shows the time histories of C_L and C_D . Table 12 shows the comparison of time-averaged C_L and C_D at different spanwise grid numbers, along with $k\omega$ SST-IDDES and URANS: γ - $k\omega$ SST. The significant improvement in the force coefficients can be clearly observed while using the transitional IDDES model over the fully turbulent IDDES model and the transitional URANS.

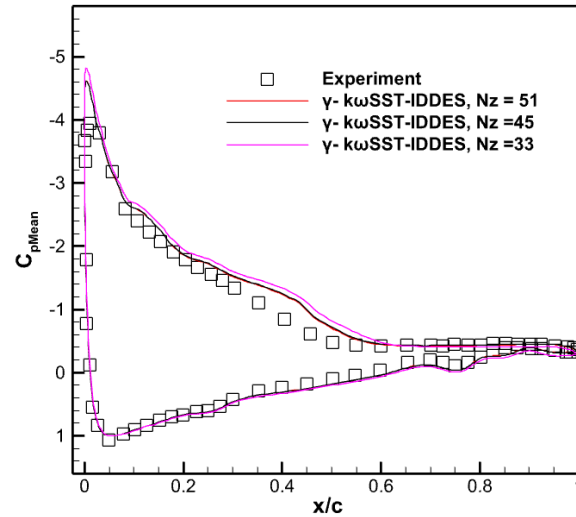


Figure 15. Distribution of the pressure coefficient C_p of the time-averaged flow field for the DBLN-526 airfoil.

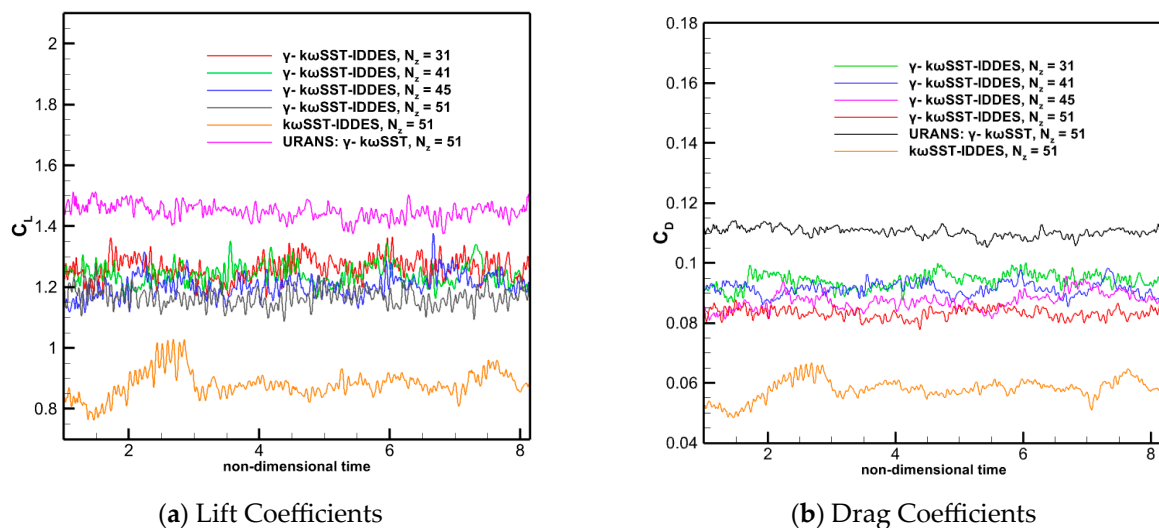


Figure 16. Time histories of lift and drag coefficients for DBLN-526 airfoil: (a) Lift Coefficients; (b) Drag Coefficients.

Table 12. Comparisons of time-averaged C_L and C_D for DBLN-526 airfoil test case.

Case	Nz	C_L	C_D
γ - $k\omega$ SST-IDDES	31	1.26405	0.10431
γ - $k\omega$ SST-IDDES	41	1.24050	0.10152
γ - $k\omega$ SST-IDDES	45	1.21155	0.09814
γ - $k\omega$ SST-IDDES	51	1.19555	0.09314
$k\omega$ SST-IDDES	51	0.87998	0.06838
URANS : γ - $k\omega$ SST	51	1.44260	0.09989
Experiment [18]	-	1.1656	0.1419

Figure 17 compares the turbulent structures of IDDES and URANS. The URANS: γ - $k\omega$ SST simulation illustrates the two-dimensional (2D) turbulent structures, whereas the IDDES produces highly complex three-dimensional (3D) structures that become slightly apparent with the increase in spanwise grid numbers. The shear layer instability induced by γ - $k\omega$ SST-IDDES appears significantly further downstream compared to that induced by $k\omega$ SST-IDDES, primarily due to the presence of a boundary layer transition. Both the γ - $k\omega$ SST-IDDES and $k\omega$ SST-IDDES predict the development of shear layer instabilities from quasi-2D large-scale structures to 3D small-scale turbulent structures.

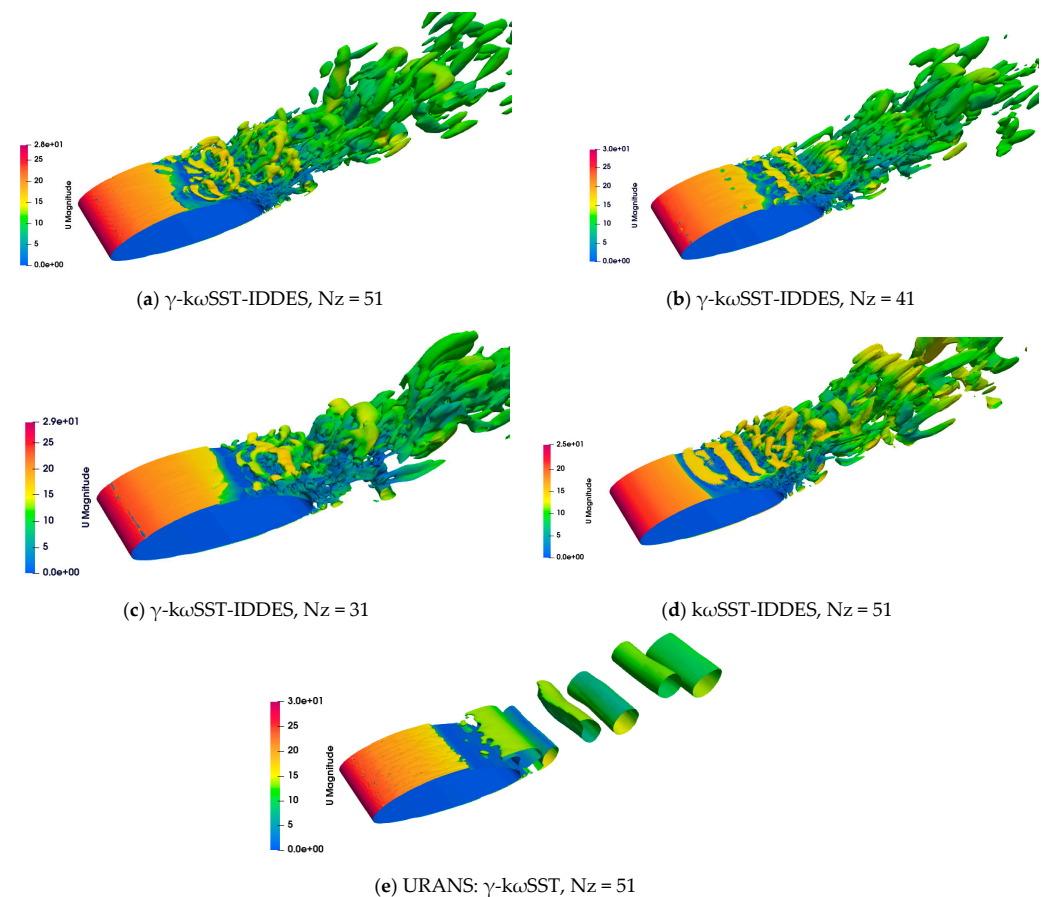


Figure 17. Instantaneous flow field results for DBLN-526 airfoil test case (Q -criterion = 300 isosurface colored by the velocity magnitude): (a) γ - $k\omega$ SST-IDDES, $N_z = 51$; (b) γ - $k\omega$ SST-IDDES, $N_z = 41$; (c) γ - $k\omega$ SST-IDDES, $N_z = 31$; (d) $k\omega$ SST-IDDES, $N_z = 51$; (e) URANS: γ - $k\omega$ SST, $N_z = 51$.

Figure 18 shows the time-averaged streamlines near the leading edge to capture the laminar separation bubble. The γ - $k\omega$ SST-IDDES model produces the laminar separation bubble with laminar separation onset at $x/c = 0.03$ and turbulent reattachment at $x/c = 0.0736$ along with the transition point near the leading edge at $x/c = 0.0567$. $k\omega$ SST-IDDES on the other hand does not give any LSB. Figure 19 shows the time-averaged streamlines near the boundary layer separation. The γ - $k\omega$ SST-IDDES model captures the flow separation bubble with onset at $x/c = 0.4400$ and reattachment at $x/c = 0.4736$. However, $k\omega$ SST-IDDES predicts no such separation bubble. Figure 20 compares the velocity profiles at different chordwise positions by the three simulation methods. The profiles from γ - $k\omega$ SST-IDDES and URANS: γ - $k\omega$ SST are quite similar, indicating that both models predict the transitional flow behavior and the LSB's extent with reasonable accuracy, primarily due to their capability to capture the laminar-to-turbulent transition effectively. In contrast, the $k\omega$ SST-IDDES method shows distinct deviations in the velocity profiles, suggesting a different representation of the boundary layer development, likely due to its

inherent approach to modeling the turbulent flow without explicitly accounting for the transitional phenomena as the γ -based models do.

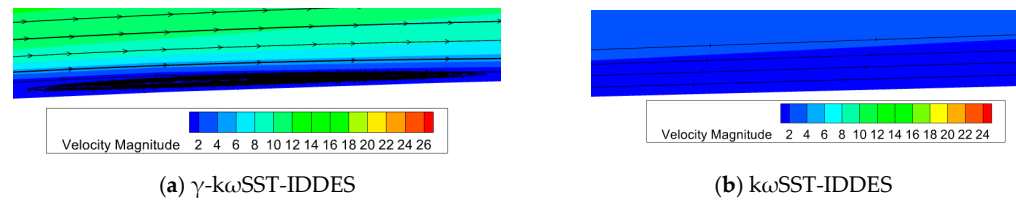


Figure 18. Time-averaged streamlines near the boundary layer separation: (a) γ -k ω SST-IDDES; (b) k ω SST-IDDES.

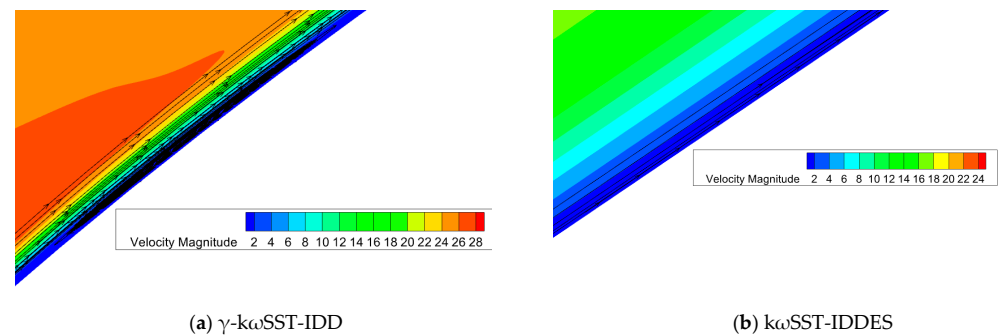


Figure 19. Time-averaged streamlines near the laminar separation bubble: (a) γ -k ω SST-IDDES; (b) k ω SST-IDDES.

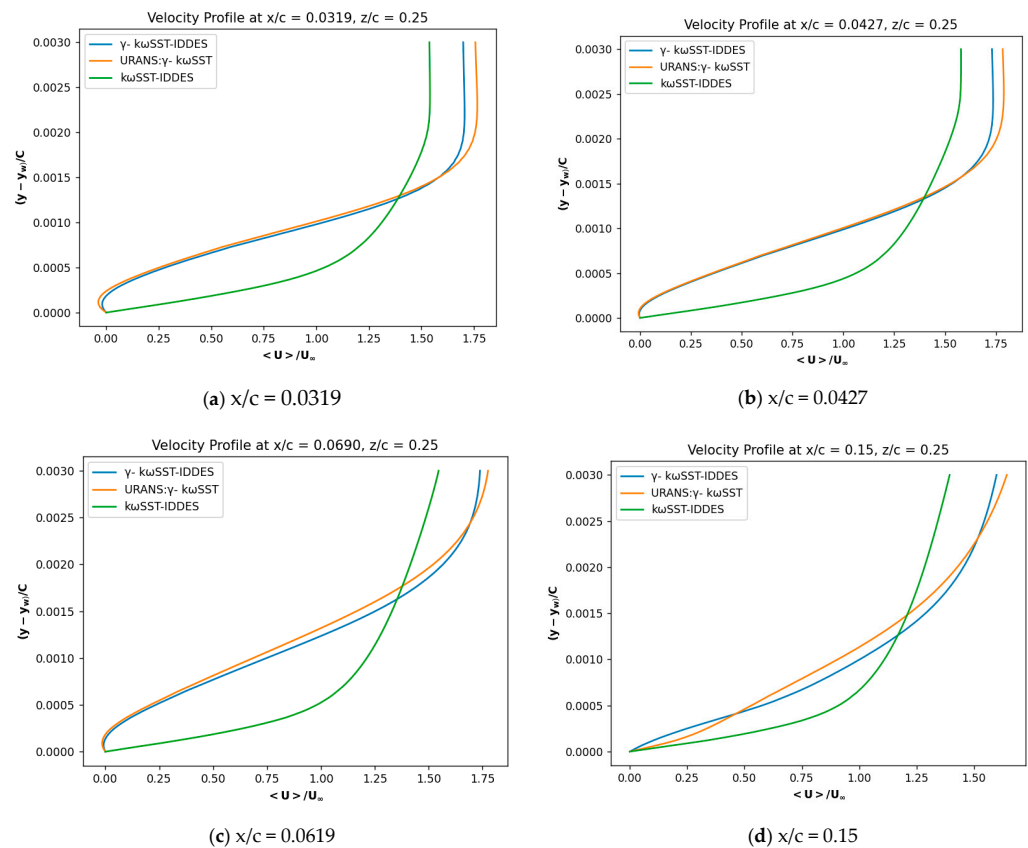


Figure 20. The profiles of mean velocity at various x/c locations: (a) $x/c = 0.0319$; (b) $x/c = 0.0427$; (c) $x/c = 0.0619$; (d) $x/c = 0.15$.

3.2.3. Flow over an NACA0021 Airfoil

A common test case for massively separated flows is the flow over the NACA0021 airfoil at an angle of attack of 60° . Wang et al. [29–31] employed this test case for fully turbulent IDDESs; however, it has not yet been utilized for transitional IDDESs.

In this study, numerical simulations are performed with a chord-based Reynolds number of 0.3×10^6 , and a turbulence intensity of 0.03%. The same “CO-grid” mesh topology as previous test cases is used for this test case in the computational domain as shown in Figure 21. A small O-grid is created with 200 points in the circumferential direction and 101 points in the normal direction, and the diameter of this O-grid is set at six times the chord length ($6c$). The height of the first grid normal to the wall is placed at $1 \times 10^{-5}c$, ensuring $y^+ < 1$. The C-grid is situated outside the O-grid and extends to $25c$ in the upstream direction and $40c$ in the downstream direction, which consists of 41 grid points upstream and 67 grid points downstream. The spanwise width is extended to $1c$, and three sets of grids with $N_z = 28$, $N_z = 40$, and $N_z = 56$ are used. A non-dimensional time step size $\Delta t U_\infty C^{-1} = 0.00016$ is used for all unsteady time-accurate computations. The computations were carried out for a total of five non-dimensional time units.

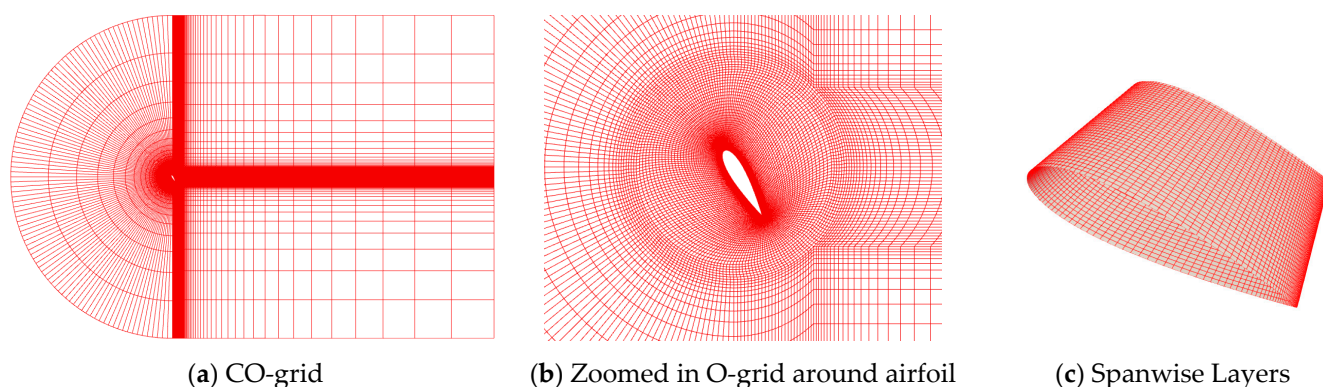


Figure 21. “CO-grid” topology around NACA0021 airfoil test case: (a) CO-grid; (b) Zoomed in O-grid around airfoil; (c) Spanwise Layers.

Figure 22 shows the distribution of the pressure coefficient of the time-averaged flow field for the NACA0021 airfoil test case at three different levels of grids. It shows a good convergence with grid refinement and gives the best agreement with experimental data [32] at $N_z = 40$. Hence, the spanwise grid with $N_z = 40$ is used for the further numerical simulations. Figure 23 shows the comparison of time histories of C_L and C_D . Table 13 shows the comparison of time-averaged C_L and C_D between IDDES and URANS simulations at different spanwise grid numbers. The γ - $k\omega$ SST-IDDES improves the lift and drag coefficients in comparison to both $k\omega$ SST-IDDES and γ - $k\omega$ SST. However, the force coefficient results obtained from URANS: γ - $k\omega$ SST largely deviate from those of the IDDES results.

Table 13. Comparisons of time-averaged C_L and C_D for NACA0021 airfoil test case.

Case	N_z	C_L	C_D
γ - $k\omega$ SST-IDDES	28	0.89218	1.46686
γ - $k\omega$ SST-IDDES	40	0.90310	1.48744
γ - $k\omega$ SST-IDDES	56	0.87306	1.43999
$k\omega$ SST-IDDES	40	0.87638	1.43950
γ - $k\omega$ SST	40	1.36597	2.34300
Experiment	-	0.92	1.52

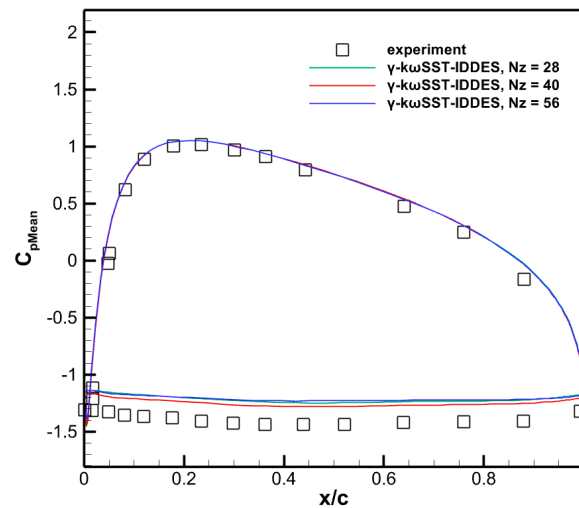


Figure 22. Distribution of the pressure coefficient C_p of the time-averaged flow field for the NACA0021 airfoil test case.

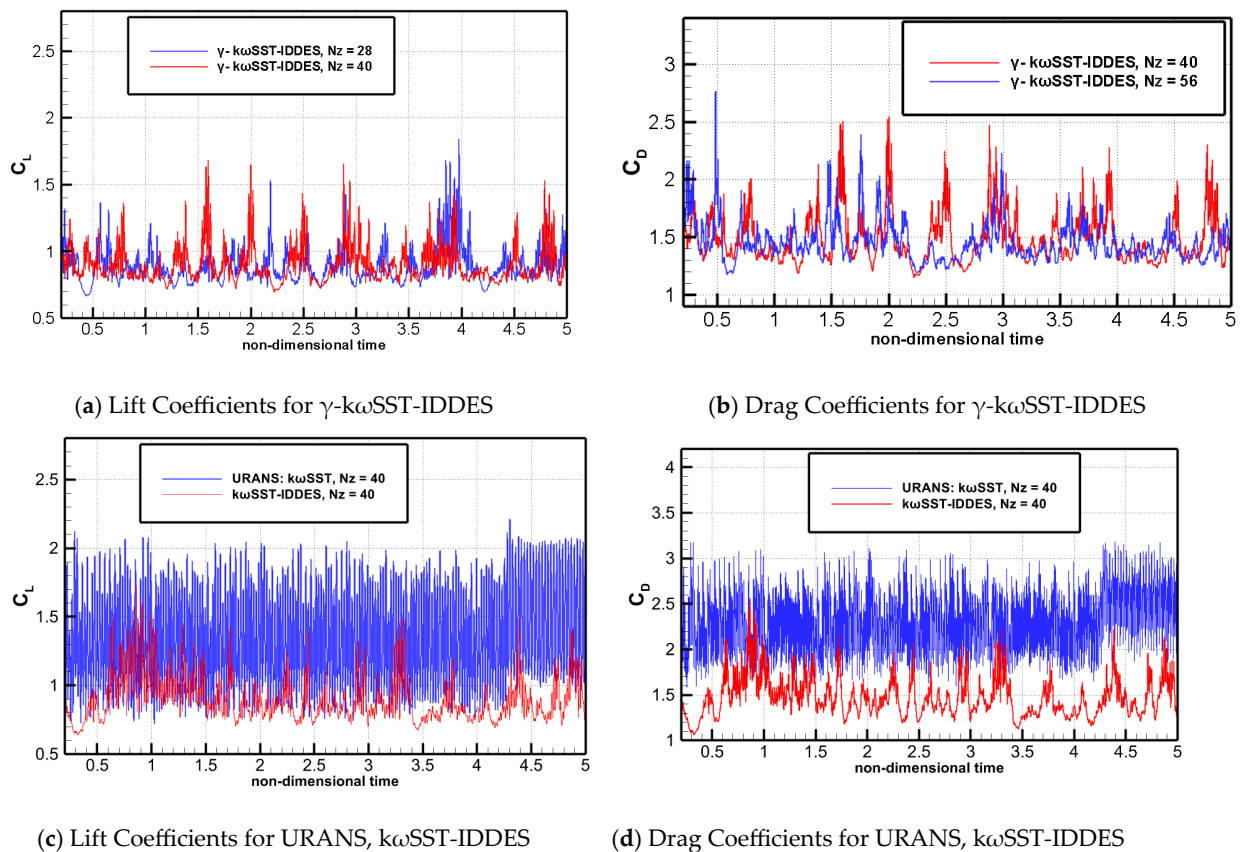


Figure 23. Time histories of force coefficients for NACA0021 airfoil test case: (a) Lift Coefficients for γ - $k\omega$ SST-IDDES; (b) Drag Coefficients for γ - $k\omega$ SST-IDDES; (c) Lift Coefficients for URANS, $k\omega$ SST-IDDES; (d) Drag Coefficients for URANS, $k\omega$ SST-IDDES.

Figure 24 shows the comparison of turbulent flow structures among the IDDES and URANS simulations. A distinction between the γ - $k\omega$ SST-IDDES and the $k\omega$ SST-IDDESs can be observed on the leading edge of the airfoil. There are fractures and the shear layer is completely detached in the leading edge of the airfoil in the $k\omega$ SST-IDDES, whereas the shear layer is attached in the γ - $k\omega$ SST-IDDES. Like the previous test cases, the URANS: γ - $k\omega$ SST simulation produces two-dimensional vortex structures. Figure 25 illustrates the time-averaged streamlines near the laminar separation bubble. The γ - $k\omega$ SST-IDDES

model yields the LSB, manifesting onset at $x/c = 0.0146$ and reattachment at $x/c = 0.0169$, alongside the transition point at $x/c = 0.0153$. This demonstrates the ability of γ - $k\omega$ SST-IDDES to effectively capture the transition properties within the flow field. Conversely, the $k\omega$ SST-IDDES model fails to characterize any LSB, highlighting the limitation of this approach in capturing certain flow phenomena.

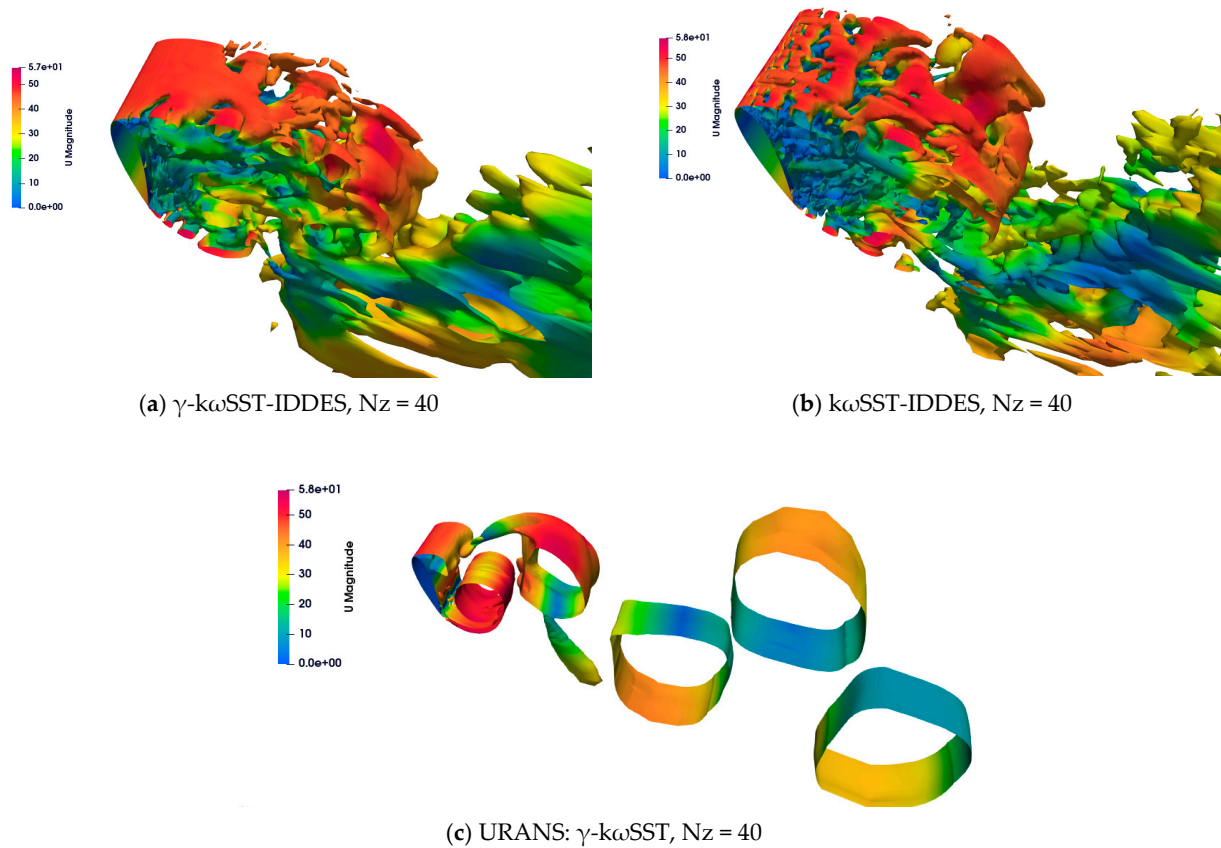


Figure 24. Instantaneous flow field results for the NACA0021 airfoil test case (Q-criterion = 500 iso-surface colored by the velocity magnitude): (a) γ - $k\omega$ SST-IDDES, $N_z = 40$; (b) $k\omega$ SST-IDDES, $N_z = 40$; (c) URANS: γ - $k\omega$ SST, $N_z = 40$.

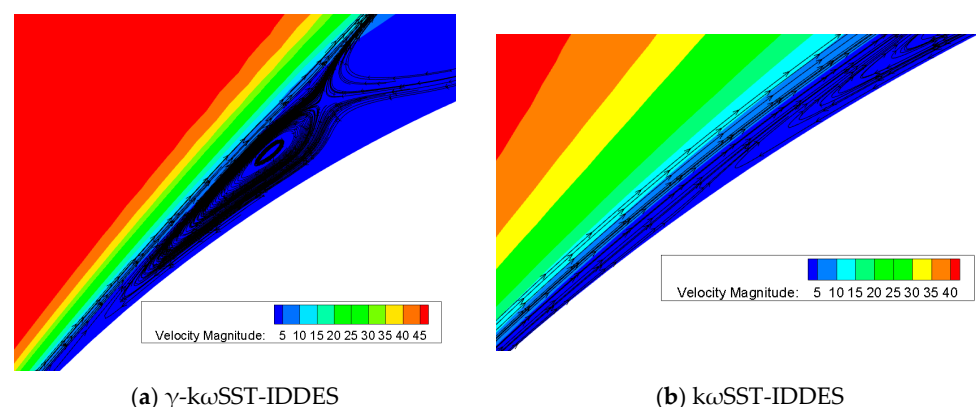


Figure 25. Time-averaged streamlines near the laminar separation bubble: (a) γ - $k\omega$ SST-IDDES; (b) $k\omega$ SST-IDDES.

4. Conclusions

In this study, the γ - $k\omega$ SST and $K\gamma$ - ω SST transitional models were newly implemented in OpenFOAM-v2312 to simulate the laminar–turbulent transition and laminar separation

bubble (LSB) around the two-dimensional E387 airfoil at angles of attack of 2° , 4° , and 6° . As the angle of attack increases, the LSB is observed to move closer to the leading edge. The size of the LSB predicted by both the γ - $k\omega$ SST and $k\omega$ SSTLM models is nearly identical at each angle of attack. Based on these results, the γ - $k\omega$ SST model was coupled with the traditional IDDES method to form a hybrid approach, referred to as γ - $k\omega$ SST-IDDES, capable of capturing both transition and large-scale separation phenomena.

To evaluate the performance of the γ - $k\omega$ SST-IDDES model, three airfoils with varying degrees of flow separation were studied: the E387 airfoil (small separation), the DBLN-526 airfoil (moderate separation), and the NACA0021 airfoil (massive separation). Across all test cases, both the γ - $k\omega$ SST-IDDES and $k\omega$ SST-IDDES models generated highly complex, three-dimensional turbulent structures, while the URANS-based γ - $k\omega$ SST simulations produced predominantly two-dimensional structures.

For the E387 airfoil, both hybrid models captured an LSB near the leading edge, although the $k\omega$ SST-IDDES predicted a significantly smaller bubble compared to γ - $k\omega$ SST-IDDES. The γ - $k\omega$ SST-IDDES predicted laminar separation at $x/c = 0.063$ and turbulent reattachment at $x/c = 0.121$. In the case of the DBLN-526 airfoil, the γ - $k\omega$ SST-IDDES accurately predicted an LSB with laminar separation at $x/c = 0.030$ and reattachment at $x/c = 0.0736$, and additionally captured a secondary separation bubble at $x/c = 0.4400$, which was not resolved by the $k\omega$ SST-IDDES model. Finally, for the NACA0021 airfoil, the γ - $k\omega$ SST-IDDES predicted an LSB with onset at $x/c = 0.0146$ and reattachment at $x/c = 0.0169$, while the $k\omega$ SST-IDDES again failed to capture this transition phenomenon.

Overall, the results demonstrate that the γ - $k\omega$ SST-IDDES model provides improved predictive capability for transitional and separated flows, especially in cases with complex laminar-to-turbulent behavior and laminar separation bubbles.

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